PRINCE NASAMU ALHASSAN

AI Researcher · Low-Resource NLP · African Language Technologies
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PROFILE

Mathematical Sciences with Computer Science student at the University of Ghana, Legon (4.0 CGPA), and independent researcher affiliated with GhanaNLP, working across low-resource NLP, speech modelling, data science, full-stack development and AI agent design. Builds open-source language and speech infrastructure for Ghanaian and African languages, and ships voice-first products for people that mainstream tools leave out, including blind users, mothers and newborns in rural communities, and students. Started with Kusaal, a severely under-resourced Gur language of Northern Ghana, where he designed, trained and publicly released translation and speech recognition models, a ~48,000-pair parallel corpus and an 81-hour speech dataset, with the full construction pipeline open-sourced. Research interests include multilingual representation learning, low-resource language modelling, speech technology for African languages, and equitable AI for African language communities.

RESEARCH EXPERIENCE

Lead Researcher, Northern Languages Sept 2026 to Present

Human-Computer Interaction (HCI) Lab, University of Ghana, Legon

- Lead research on speech and language technology for the languages of Northern Ghana.
- Developing a research plan for a community-led wearable platform that collects consented, naturalistic everyday speech for African languages, with community-led transcription and validation, participant withdrawal and deletion rights, and community governance over data access.

Machine Translation and Speech Recognition for Low-Resource Ghanaian Languages (Kusaal) 2024 to Present

Independent Research | Affiliated with GhanaNLP

- Curated and openly released a Kusaal-English parallel corpus, grown from 32,589 to roughly 48,000 sentence pairs across scripture, lexicographic and encyclopedic registers, published with the full construction pipeline under CC BY 4.0.
- Fine-tuned Meta's NLLB-200-distilled-600M in both directions, reaching BLEU 27.57 (Kusaal to English) and 13.72 (English to Kusaal) on the in-domain held-out test set.
- Designed a three-stage corpus-mining pipeline over Kusaal Wikipedia: anchor alignment on shared digits and proper names, MT-bridged semantic verification using mpnet cosine similarity and chrF++, and manual adjudication of 503 borderline pairs. Yielded 13,659 verified pairs (8,268 gold, 5,391 silver) from 1,170 comparable article pairs, moving the corpus beyond its original scripture-heavy register.
- Retrained on the combined 86,683-example set with full legacy replay, lifting scores on a frozen 1,000-sentence Kusaal Wikipedia benchmark from 42.17 to 47.73 BLEU (Kusaal to English) and 19.43 to 32.26 BLEU (English to Kusaal), with in-domain performance held steady. The 1,000-sentence benchmark is released publicly as a separate dataset.
- Built and released a Kusaal ASR dataset of 30,820 verse-level clips (81.7 hours, 16 kHz mono) with book-level held-out splits, and trained a Whisper-small LoRA model reaching 30.41% WER on unseen books.
- Published models, datasets and working demo Spaces on Hugging Face (kusaal-nllb-600M, kusaal-whisper-small-lora, kusaal-english-parallel-corpus, kusaal-asr-dataset); code and documentation at github.com/NasamuAlhassan/kusaal-mt.
- Preparing three papers with Dr. Anthony Agoswin Musah (University of Education, Winneba; author of the Kusaal reference grammar and co-author of the current orthography): a dataset paper, a machine translation paper and a corpus-construction methodology paper.
- Presented "Bridging the Gap: A Fine-Tuned NLLB-200 Model for Kusaal-English Machine Translation" at the Google Research Africa Monthly Student Tech-Talk, AICC Accra (31 July 2026), with positive feedback from Google researchers, and at a GhanaNLP community session.

Multilingual Speech Modelling for Ghanaian Languages 2026

GhanaNLP TTS/ASR Hackathon | AI Skills and Compute Africa (AISCA) H200 compute window

- Trained KASA-42, a multilingual ASR model over the 42-language ghana-speech corpus (w2v-BERT 2.0 with a CTC head, bf16), reaching 30.2% WER on a book-disjoint test set and 96.8% language-identification accuracy, prioritising northern Ghanaian languages.
- Identified narrator leakage between splits in scripture-sourced audio and redesigned the evaluation with source-disjoint splits, reporting a leaked-versus-honest WER comparison as part of the submission.
- Trained a Kusaal text-to-speech model in the same 48-hour window; delivered ONNX exports and a Gradio demo, and contributed to the ghananlpcommunity Hugging Face organisation.

EDUCATION

University of Ghana, Legon 2025 to 2029 (Expected)

BSc Mathematical Sciences with Computer Science | CGPA: 4.0 / 4.0

- Relevant coursework: Calculus, Linear Algebra, Probability and Statistics, Discrete Mathematics, Algorithms, Data Structures, Numerical Analysis.

Osei Tutu Senior High School, Akropong Completed 2025

WASSCE: 6 A1s, 2 B3s | Ranked 1st in class, 5th school-wide | Highest Elective Mathematics score in cohort

- National Science and Maths Quiz (NSMQ) national semi-finalist; team leader and primary organiser for the school's national campaign.
- Awarded Best Student, with recognition for top performance in Mathematics, Physics, Chemistry and Biology.

SELECTED PROJECTS

Tɛkyerɛma AI: Voice-First Android Assistant for Blind and Low-Vision Users 2026

MTN Ghana Tɛkyerɛma Pa Hackathon (MTN Ghana and University of Ghana HCI Lab) | Top five finalist

- Built an Android app that lets blind and low-vision users handle mobile money, messaging and calling entirely by voice, in Ghanaian English and Asante Twi, including transfers to other networks.
- Evolved from the Zigi Voice prototype, a finite-state agentic architecture using Khaya API Twi ASR and TTS against MTN MoMo transaction flows; tuned to run on a low-end phone (Android 11, about 2.8 GB RAM).

Mabia AI: Voice-First Maternal and Newborn Care Coordination 2026

UNICEF StartUp Lab "AI for Nurturing Care" Hackathon | Engineering and AI lead, team of three

- Built an IVR and USSD platform for CHPS-level maternal and child health in Northern Ghana: proactive voice outreach on the WHO eight-contact schedule, a deterministic and auditable triage engine returning readable reason codes, keypad-based dietary-diversity screening with seasonal and affordability filtering, and community transport dispatch with referral tracking to receipt of care.
- React PWA with IndexedDB offline capture, FastAPI backend over an append-only event log with client-generated idempotency keys, PostgreSQL, Africa's Talking voice/SMS/USSD, Docker; 236 automated tests. Live at mabia-ai.vercel.app.
- Local-language voice layer in Dagbani and Kusaal via the Khaya API and Meta MMS, with recorded and spoken-English fallbacks.

GreenReS: Climate Risk Education and Action Platform 2026

GreenRes Hackathon 2026, Africa Climate Collaborative and Mastercard Foundation | Technical lead, team of three

- Shipped a React PWA and FastAPI backend delivering district-level climate risk guidance to rural users over SMS and USSD, with a three-tier proof model for verifying real-world action; 24 endpoints, 76 tests. Live at green-res.vercel.app.

Ghana AI-NCD Navigator: National Non-Communicable Disease Platform 2026

- Building a DHIS2-based national platform for non-communicable diseases with an AI risk layer for conditions such as hypertension, diabetes and cardiovascular disease.

Wake the Word 2026

Gloo Hackathon (YouVersion Platform) | Solo build

- Designed and built a Scripture-driven app for the hackathon on the YouVersion Platform.

RBAS: Real-Time Biotech Regulatory Application System 2026

Client project for Ghana's National Biosafety Authority (NBA) | Sole developer

- Building an online biosafety permit application system proposed to the NBA, delivered end to end as a working system, with a demonstration requested by the Authority.

CampusLink: Verified Student Connection Platform 2025 to Present

- Founded and built a platform for African university students (Next.js 14, Supabase, Gemini API) with two-stage student identity verification, real-time messaging and an interactive campus map; live at campus-link-sigma.vercel.app.

Kasa: Voice-First English and Twi Language Coaching 2026

- Founded and specified a voice-first coaching app for Twi speakers who need working English (React Native/Expo, Supabase, Azure Pronunciation Assessment), with a full product and technical requirements document.

Forge: AI-Powered Academic Productivity Tool May 2026

DesignPxD Student AI Hackathon | University of Ghana, Legon

- Voice-driven study scheduler (React/Vite, Supabase, Vercel, Gemini 2.5 Flash) with Pomodoro focus sessions, usage analytics, PDF/ICS export and a speech-to-speech assistant panel; built and shipped within 48 hours.

ShieldNet: Cybersecurity Threat Intelligence Platform 2026 (in development)

- Architecting a multi-tier cybersecurity SaaS with a team: AI-powered threat intelligence engine, mobile money payment integration and emergency SMS notification infrastructure.

Earlier Work 2025 to 2026

- Mabia Voice Line, submitted to SafeTech Africa HackLab 2026 (UNFPA and Spotlight Initiative).
- Nasamu, a personal AI workspace over a self-hosted open-weight language model running on free GPU compute.
- FieldMind, a retrieval-augmented generation tool for enterprise document search built on the Gemini API (LabLab.ai submission); HyperFlow Risk Agent, a team entry in the AMD Developer Cloud Hackathon AI Agents track; WhatsApp and Telegram AI agents, including a locally hosted Telethon and Ollama conversational agent.

PROFESSIONAL EXPERIENCE

Technology and Digital Systems Intern, EGA Mentorship International 2026

- Built and shipped projects across EGA's digital systems, completed a technical audit of the organisation's web properties and presented the findings in a PowerPoint deck, and mentored junior team members.

- Led the backend (Supabase: Postgres, auth, storage, REST API) of a blog-style website with admin portal for the Dr. Kwame Nkrumah Political Youth Fun Club, working with a frontend team.

Data Science Intern, Decodelabs April to May 2026

- Built an end-to-end climate data pipeline ingesting multi-city meteorological time series from the Open-Meteo API across five Ghanaian cities, producing structured datasets for downstream analytics.

Data Science Intern, Codveda Technologies 2025

- Executed a full data science pipeline: web-scraped 1,000+ records with BeautifulSoup, performed data cleaning, built regression and classification models, applied ARIMA time-series forecasting, and developed NLP text classification systems.

AWARDS, SCHOLARSHIPS, CERTIFICATIONS AND PROGRAMS

- Top five finalist, MTN Ghana Tɛkyerɛma Pa Hackathon 2026.
- MTN Ghana Foundation Bright Scholarship, 2026 cohort.
- Anthropic Claude for Open Source Program, awarded 2026 (six months of Claude Max).
- Official Ambassador, International Organization of Youth (IOY), 2026 to 2027.
- Columbia University (Columbia+), Prompt Engineering & Programming with OpenAI, April 2026.
- TechCrush Data Science Bootcamp Scholarship, 15-week applied curriculum with live instruction; completed August 2026, accredited by the American Council of Training and Development, USA.
- Goalpha Intern Mentorship Program (GIMP), participant.
- SYNC (Ckodon Foundation) mentorship program, accepted.
- Participant, Pan African AI & Innovation Summit 2026.
- British Airways virtual job simulation (Forage) and GreenRes Bootcamp certificate.

LEADERSHIP AND COMMUNITY

- Campus Ambassador, WIWIK 100, University of Ghana.
- Active building member, Claude Builders Club, University of Ghana; led "Team Gold" in the chapter hackathon.
- Contributor to GhanaNLP, the African NLP research network, and member of COMPSSA, the Computer Science Students' Association at the University of Ghana.
- Self-directed learner: taught himself programming and graphic design during the COVID-19 lockdowns, before any formal computer science instruction.
- Church leadership roles: drummer, sound engineer, choir member, event organiser and chaplain.

TECHNICAL SKILLS

- **Languages and Frameworks:** Python, JavaScript/TypeScript, React, Next.js 14, React Native/Expo, Android, FastAPI, Node.js, SQL
- **ML, NLP and Speech:** PyTorch, Hugging Face Transformers, PEFT/LoRA, NLLB-200, Whisper, w2v-BERT 2.0, CTC with KenLM decoding, Meta MMS, sentence-transformers, scikit-learn, pandas, RAG systems, ONNX, Gradio
- **Infrastructure and Data:** Supabase, PostgreSQL, Vercel, Docker, Git/GitHub, Kaggle, Google Colab, Lightning AI, Jupyter, Africa's Talking, Khaya API, API integration
- **Human Languages:** Kusaal (native), English (native)